Applications: Growth and Decay

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Growth models**

A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.

a State the initial population and the annual growth rate. Initial population 12,000; growth rate 3.5% per year.

b Find the population after 10 years, to the nearest person. $P(10) = 12000(1.035)^{10} \approx 16{,}927$.

(a) Initial population 12,000; growth rate 3.5% per year.; (b) $P(10) = 12000(1.035)^{10} \approx 16{,}927$.

2 **[Foundational]** • **Target: Growth models**

An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.

$2 = e^{0.04t}$, so $t = \frac{\ln 2}{0.04} \approx 17.33$ years.

3 **[Foundational]** • **Target: Decay models**

Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.

a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places. $\left(\frac{1}{2}\right)^{2000/5730} \approx 0.785$.

b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years. $0.3 = (0.5)^{t/5730}$ gives $t = 5730\,\frac{\ln 0.3}{\ln 0.5} \approx 10{,}000$ years.

(a) $\left(\frac{1}{2}\right)^{2000/5730} \approx 0.785$.; (b) $0.3 = (0.5)^{t/5730}$ gives $t = 5730\,\frac{\ln 0.3}{\ln 0.5} \approx 10{,}000$ years.

4 **[Foundational]** • **Target: Decay models**

A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.

a Find the decay constant k (4 d.p.). $\frac{50}{200} = e^{12k}$, so $k = \frac{\ln(1/4)}{12} \approx -0.1155$.

b Find the half-life of the substance. $200 \to 50$ is two halvings in 12 h, so the half-life is exactly 6 hours (equivalently $\frac{\ln 2}{0.1155} \approx 6.0$ h).

(a) $\frac{50}{200} = e^{12k}$, so $k = \frac{\ln(1/4)}{12} \approx -0.1155$.; (b) $200 \to 50$ is two halvings in 12 h, so the half-life is exactly 6 hours (equivalently $\frac{\ln 2}{0.1155} \approx 6.0$ h).

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5 [Foundational] • Target: Decay models

Newton's law of cooling: coffee at $90°$C cools in a $20°$C room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.

$T(10) = 20 + 70e^{-1} \approx 20 + 25.75 = 45.8°$C.

6 [Foundational] • Target: Logistic growth (an introduction)

A lake is stocked with fish, and the population is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$, where t is in years.

a Find the initial population $P(0)$. $P(0) = \frac{5000}{1+9} = 500$.

b Find the carrying capacity (the value P approaches as $t \to \infty$). As $t \to \infty$, $e^{-0.4t} \to 0$, so $P \to \frac{5000}{1} = 5000$.

c Find $P(10)$, to the nearest fish. $P(10) = \frac{5000}{1 + 9e^{-4}} \approx \frac{5000}{1.1648} \approx 4292$.

(a) $P(0) = \frac{5000}{1+9} = 500$.; (b) As $t \to \infty$, $e^{-0.4t} \to 0$, so $P \to \frac{5000}{1} = 5000$.; (c) $P(10) = \frac{5000}{1 + 9e^{-4}} \approx \frac{5000}{1.1648} \approx 4292$.

7 [Foundational] • Target: Comparing models

Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.

Town A: $10000 + 300(15) = 14{,}500$. Town B: $10000(1.03)^{15} \approx 10000(1.5580) \approx 15{,}580$. Town B is larger.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing models

A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model $A(t) = 80e^{kt}$, finding k if the total elapsed time is 15 years (4 d.p.).

$80 \to 10$ over 3 halvings ($80 \to 40 \to 20 \to 10$) means 3 half-lives $= 15$ years, so half-life $= 5$ years. Using $A(15) = 10$: $10 = 80e^{15k}$, so $k = \frac{\ln(1/8)}{15} \approx -0.1386$.

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9 [Intermediate] • Target: Visualizing the logistic curve

The fish population model $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$ (from earlier) is graphed below. Use the graph to estimate where the population is growing fastest, then explain why this happens near the inflection point.

The curve is steepest near its midpoint, roughly where $P \approx 2500$ (half the carrying capacity), consistent with the logistic model's fastest growth occurring at $P = \frac{L}{2}$.

10 [Intermediate] • Target: Doubling time and half-life formulas

Derive a general formula for the doubling time T of a quantity growing as $A(t) = A_0 e^{kt}$, in terms of k.

Set $2A_0 = A_0 e^{kT}$: $2 = e^{kT} \Rightarrow \ln 2 = kT \Rightarrow T = \frac{\ln 2}{k}$.

11 [Intermediate] • Target: Doubling time and half-life formulas

Using the doubling-time formula derived above, find the doubling time for a population growing at a continuous rate of $k = 0.025$ per year, to one decimal place.

$T = \frac{\ln 2}{0.025} \approx \frac{0.6931}{0.025} \approx 27.7$ years.

12 [Intermediate] • Target: Half-life formula derivation

Derive a general formula for the half-life $T_{1/2}$ of a quantity decaying as $A(t) = A_0 e^{kt}$ (with $k < 0$), in terms of k.

Set $\frac{A_0}{2} = A_0 e^{kT_{1/2}}$: $\frac{1}{2} = e^{kT_{1/2}} \Rightarrow \ln\left(\frac{1}{2}\right) = kT_{1/2} \Rightarrow T_{1/2} = \frac{-\ln 2}{k}$ (positive since $k < 0$).

13 [Intermediate] • Target: Half-life formula derivation

A radioactive isotope has decay constant $k = -0.023$ per year. Use the formula above to find its half-life, to one decimal place.

$T_{1/2} = \frac{-\ln 2}{-0.023} = \frac{0.6931}{0.023} \approx 30.1$ years.

14 [Intermediate] • Target: Combining growth models with rates

A savings account earns continuous interest, growing from 5000 to 5512.50 riyals in 2 years. Find the annual interest rate r using $A = Pe^{rt}$, to two decimal places (as a percentage).

$5512.50 = 5000e^{2r} \Rightarrow 1.1025 = e^{2r} \Rightarrow 2r = \ln(1.1025) \approx 0.0976 \Rightarrow r \approx 0.0488$, i.e. approximately 4.88%.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The half-life of Carbon-14 is 5730 years. (a) Write $A(t)$. (b) How much of 200 g remains after 11 460 years? (c) When is only 10% left?

(a) Exponential Decay Function: $A(t) = 200\left(\frac{1}{2}\right)^{t/5730}$. (b) Mass Remaining at $t = 11{,}460$ years: Notice that $\frac{11{,}460}{5730} = 2$ half-lives: $A(11{,}460) = 200\left(\frac{1}{2}\right)^2 = 200\left(\frac{1}{4}\right) = \mathbf{50}$ g. (c) Time to Reach 10% of Initial Mass: Set $\frac{A(t)}{200} = 0.10$.

16 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 3.03 Sum and Difference Formulas

Level 2 · MediumA table of values for a positive function $g(x)$ over equally spaced inputs $x = 0, 1, 2, 3$ produces outputs $g(0) = 4, g(1) = 12, g(2) = 36, g(3) = 108$. Which function family best models $g(x)$, and what is the constant growth factor per unit increase in x?

Calculate the consecutive ratios of consecutive outputs: $\frac{g(1)}{g(0)} = \frac{12}{4} = 3, \quad \frac{g(2)}{g(1)} = \frac{36}{12} = 3, \quad \frac{g(3)}{g(2)} = \frac{108}{36} = 3.$ Because the ratio of outputs is constant for equal-length input intervals, $g(x)$ is an exponential function with a constant growth factor of 3.

17 [Challenging] • Target: Growth models • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

A stone is thrown vertically upwards with speed 15 m/s from the top of a cliff 40 m above sea level. Taking $g = 9.8$ m/s^2 and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

(a) At maximum height, $v = 0$. $u = +15$ m/s, $a = -9.8$ m/s^2: $v^2 = u^2 + 2as \Rightarrow 0 = 15^2 + 2(-9.8)s \Rightarrow 19.6s = 225 \Rightarrow s = \frac{225}{19.6} \approx 11.48 \text{ m} \approx 11.5 \text{ m}$ (b) At the sea, the displacement relative to the launch point is $s = -40$ m. Use $s = ut + \frac{1}{2}at^2$: $-40 = 15t + \frac{1}{2}(-9.8)t^2 \Rightarrow -40 = 15t - 4.9t^2 \Rightarrow 4.9t^2 - 15t - 40 = 0$ Solve via quadratic formula.

18 [Challenging] • Target: Decay models • Cross-Lesson: 4.01 Circles and Parabolas

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

Rewrite in index form: $y = 4x^{-1/2} + 2x$. At $x = 4$: $y = 4(4^{-1/2}) + 2(4) = 4\left(\frac{1}{2}\right) + 8 = 2 + 8 = 10$. Point of contact is $(4, 10)$. Differentiate: $\frac{dy}{dx} = 4\left(-\frac{1}{2}x^{-3/2}\right) + 2 = -2x^{-3/2} + 2 = -\frac{2}{x^{3/2}} + 2$ At $x = 4$: $x^{3/2} = (\sqrt{4})^3 = 8$. Thus: $m_{\tan} = -\frac{2}{8} + 2 = -\frac{1}{4} + 2 = \frac{7}{4}$ The normal has gradient $m_{\text{norm}} = -\frac{1}{7/4} = -\frac{4}{7}$. Using point $(4, 10)$.

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19 [Challenging] • Target: Logistic growth (an introduction) • Cross-Lesson: 4.02 Ellipses

A Ferris wheel has a diameter of 40 meters and its lowest point is 4 meters above the ground. Passengers board at the lowest point at $t = 0$, and the wheel completes one full revolution every 8 minutes. Write a cosine model $h(t)$ for a passenger's height at time t, and find the times during the first revolution when $h(t) = 34$ meters.

Step 1: Determine sinusoidal parameters: Radius (amplitude) $= 20$ m; midline $D = 4 + 20 = 24$ m; angular frequency $B = \frac{2\pi}{8} = \frac{\pi}{4}$. Starting at the minimum gives a negative cosine: $\mathbf{h(t) = -20\cos\left(\frac{\pi}{4}t\right) + 24}$. Step 2: Solve $h(t) = 34$ on $[0, 8]$: $-20\cos\left(\frac{\pi}{4}t\right) + 24 = 34 \Rightarrow \cos\left(\frac{\pi}{4}t\right) = -\frac{1}{2}$. .

20 [Challenging] • Target: Comparing models • Cross-Lesson: 4.03 Hyperbolas

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

Separate variables and integrate: $\int \frac{1}{\theta - 20}\, d\theta = -\int k\, dt \Rightarrow \ln(\theta - 20) = -kt + C$ Apply initial condition $t = 0, \theta = 80$: $\ln(60) = C \Rightarrow \ln\left(\frac{\theta - 20}{60}\right) = -kt \Rightarrow \theta - 20 = 60e^{-kt}$ Apply condition $t = 10, \theta = 50$: $30 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{10} \approx 0.06931$ Set $\theta = 30$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Growth models

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.

a. 8 **b.** 2

c. 4 ✓ **d.** 10

Applying Exponential & Logarithmic Functions principles, $2^{10/5} = 2^2 = 4$. Therefore, the verified correct option is 4.

22 [Foundational] • Target: Decay models

For $A = A_0 e^{-kt}$, find the half-life $t_{1/2}$ in terms of k.

a. $\frac{k}{\ln 2}$ **b.** $\frac{\ln 2}{k}$ ✓

c. $\frac{1}{k}$ **d.** e^k

$1/2 = e^{-kt_{1/2}} \Rightarrow \ln(1/2) = -kt_{1/2} \Rightarrow t_{1/2} = \ln 2/k$.

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23 [Foundational] • Target: Logistic growth (an introduction)

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

a. **10** ✓

b. 15

c. 20

d. 60

The problem requires us to find the number of combinations of selecting 3 people from a group of 5. This is a direct application of the combination formula ${}^{n}C_{r}$, where n is the total number of items to choose from, and r is the number of items to choose. In this case, $n = 5$ and $r = 3$. Using the formula: ${}^{5}C_{3} = \frac{5!}{3!\,(5-3)!} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times (2 \times 1)} = \frac{5 \times 4}{2} = \frac{20}{2} = 10$. Therefore, there are 10 different committees that can be forme.

24 [Foundational] • Target: Comparing models

A sequence is defined by the recursive formula $a_n = 2a_{n-1} + 1$, with the first term $a_1 = 3$. What is the value of the third term, a_3?

a. 7

b. 10

c. **15** ✓

d. 21

The recursive formula for the sequence is $a_n = 2a_{n-1} + 1$, and the first term is $a_1 = 3$. To find the second term, a_2, we substitute $n = 2$ into the formula: $a_2 = 2a_{2-1} + 1 = 2a_1 + 1$ Since $a_1 = 3$, we have: $a_2 = 2(3) + 1 = 6 + 1 = 7$ To find the third term, a_3, we substitute $n = 3$ into the formula: $a_3 = 2a_{3-1} + 1 = 2a_2 + 1$ Since $a_2 = 7$, we have: $a_3 = 2(7) + 1 = 14 + 1 = 15$

25 [Foundational] • Target: Visualizing the logistic curve

Consider the arithmetic sequence where the first term $a_1 = 7$ and the common difference $d = -3$. Which of the following is the explicit formula for the nth term of this sequence?

a. $a_n = 7 + 3(n-1)$

b. $a_n = 7 - 3(n-1)$ ✓

c. $a_n = 3 - 7(n-1)$

d. $a_n = -3 + 7(n-1)$

An arithmetic sequence is defined by the formula $a_n = a_1 + (n-1)d$, where a_n is the nth term, a_1 is the first term, and d is the common difference. In this problem, we are given that the first term $a_1 = 7$ and the common difference $d = -3$. Substituting these values into the formula, we get $a_n = 7 + (n-1)(-3)$, which can be rewritten as $a_n = 7 - 3(n-1)$.

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26 [Foundational] • Target: Doubling time and half-life formulas

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

a. Cannot be proved simply

b. By FTA

c. **By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation ✓**

d. Use telescoping

Applying Sequences & Series principles, Standard induction. Therefore, the verified correct option is By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation.

27 [Foundational] • Target: Half-life formula derivation

Find the closed form for F_n (Fibonacci) using Binet's formula.

a. $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ **where** $= 1 + 52, = 1 - 52$ ✓

b. $F_n = \phi^{n-1}$

c. $F_n = 2^n/5$

d. $F_n = n\phi$

Applying Sequences & Series principles, Binet's formula. Therefore, the verified correct option is $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ where $= 1 + 52, = 1 - 52$.

28 [Foundational] • Target: Combining growth models with rates

The binomial probability formula is:

a. $P(X = k) = np^k$

b. $P(X = k) = p^k(1-p)^{n-k}$

c. $P(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$ ✓

d. $P(X = k) = \binom{n}{k}p^n$

Applying Probability & Statistics principles, Binomial distribution formula. Therefore, the verified correct option is $P(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Growth models**

Derive the formula for the survival function of the exponential distribution.

a. $P(X > t) = \lambda e^{-\lambda t}$

b. $P(X > t) = e^{-\lambda t}$ ✓

c. $P(X > t) = 1 - \lambda t$

d. $P(X > t) = 1 - e^{-\lambda t}$

Applying Probability & Statistics principles, $P(X > t) = 1 - F(t) = e^{-\lambda t}$. Therefore, the verified correct option is $P(X > t) = e^{-\lambda t}$.

30 **[Intermediate]** • **Target: Decay models**

The formula for the sum of n terms of an arithmetic series is:

a. $S_n = \frac{a_1}{1-r}$

b. $S_n = \frac{n}{2}(a_1 + a_n)$ ✓

c. $S_n = na_1$

d. $S_n = a_1 \cdot \frac{r^n - 1}{r - 1}$

Applying Sequences & Series principles, Arithmetic sum formula. Therefore, the verified correct option is $S_n = \frac{n}{2}(a_1 + a_n)$.

31 **[Intermediate]** • **Target: Logistic growth (an introduction)**

Formulate the null and alternative hypotheses for the following scenario: "A company claims its average delivery time is 3 days. A customer believes it takes longer."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$ ✓

Applying Hypothesis Testing principles, $H_0 : \mu = 3,\ H_1 : \mu > 3$. Therefore, the verified correct option is $H_0 : \mu = 3,\ H_1 : \mu > 3$.

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32 [Intermediate] • Target: Comparing models

Balls are drawn one at a time, without replacement, from a bag of 20 balls; X is the number of red balls drawn. Is X binomially distributed?

a. Yes — but only if $n > 30$

b. Yes — fixed, independent trials with constant success probability

c. **No — trials are not independent / probability is not constant ✓**

d. No — the number of trials is not fixed

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

33 [Intermediate] • Target: Visualizing the logistic curve

A curve satisfies $x^2 + y^2 = 25$. Find $\frac{dy}{dx}$ at the point $(3, -4)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$ ✓

c. $\frac{4}{3}$

d. $-\frac{4}{3}$

Differentiating implicitly: $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$. At $(3, -4)$: $\frac{3}{4}$.

34 [Intermediate] • Target: Doubling time and half-life formulas

The formula for the nth term of an arithmetic sequence with first term a and common difference d is:

a. $a_n = a + (n-1)d$ ✓

b. $a_n = nd$

c. $a_n = a \cdot d^{n-1}$

d. $a_n = a + nd$

Applying Sequences & Series principles, Standard formula. Therefore, the verified correct option is $a_n = a + (n-1)d$.

35 [Intermediate] • Target: Half-life formula derivation

From a cumulative frequency curve with $n = 80$, the median is read at cumulative frequency:

a. 20

b. 40 ✓

c. 60

d. 80

Applying Representation of Data principles, Median at $n/2 = 40$ Therefore, the verified correct option is 40.

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36 [Intermediate] • Target: Combining growth models with rates

A curve is defined parametrically by $x = 2t^2$, $y = 3t^3$. Find $\frac{dy}{dx}$ at $t = 1$.

a. $\frac{7}{3}$

b. $\frac{9}{4}$ ✓

c. $-\frac{9}{4}$

d. $\frac{4}{9}$

$\frac{dx}{dt} = 4t$, $\frac{dy}{dt} = 9t^2$. $\frac{dy}{dx} = \frac{9t^2}{4t} = \frac{9t}{4}$. At $t = 1$: $\frac{9}{4}$.

37 [Intermediate] • Target: Growth models

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = t^2 - 4t$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s (excluding $t = 0$)

b. $t = 5$ s (excluding $t = 0$)

c. $t = 0$ s only

d. $t = 4$ s (excluding $t = 0$) ✓

Setting $v = 0$: $t^2 - 4t = 0 \Rightarrow t(t-4) = 0 \Rightarrow t = 4$ s (excluding $t = 0$)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Decay models • Cross-Lesson: 4.01 Circles and Parabolas

State Euler's formula and use it to find $e^{i\pi} + 1$.

a. 2

b. e^{π}

c. -2

d. 0 **(Euler's identity $e^{i\pi} + 1 = 0$)** ✓

Euler: $e^{i\theta} = \cos\theta + i\sin\theta$. At $\theta = \pi$: $e^{i\pi} = -1 + 0i = -1$.

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39 [Challenging] • Target: Logistic growth (an introduction) • Cross-Lesson: 4.02 Ellipses

A ball is dropped from 10 m and bounces to 80% each time. Total distance (all bounces, both up and down)?

a. 100 m

b. 80 m

c. 50 m

d. 90 **m** ✓

Downward: 10 m. Up and down: $2 \cdot 10 \cdot 0.8/(1 - 0.8) = 80$. Total: $10 + 80 = 90$ m.

40 [Challenging] • Target: Comparing models • Cross-Lesson: 4.03 Hyperbolas

Does the point $(4, 0)$ satisfy the implicitly defined curve $x^2 + y^2 = 16$?

a. No, the point does not lie on the curve

b. The equation does not define a curve

c. **Yes, the point lies on the curve** ✓

d. Only if r is negative

Substitute: $(4)^2 + (0)^2 = 16$, compare to 16: equal, so it is on the curve.

41 [Challenging] • Target: Visualizing the logistic curve • Cross-Lesson: 5.01 Vectors: Components and Operations

Show that $e^{i\pi} + 1 = 0$ using Euler's formula.

a. $e^{i\pi} = i$

b. $e^{i\pi} = \cos\pi + i\sin\pi = -1 \Rightarrow e^{i\pi} + 1 = 0$ ✓

c. $e^{i\pi} = 1$

d. Euler's formula is $e^{ix} = \sin x + i\cos x$

Euler: $e^{i\theta} = \cos\theta + i\sin\theta$. At $\theta = \pi$: $-1 + 0i = -1$.

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42 [Challenging] • Target: Doubling time and half-life formulas • Cross-Lesson: 5.02 The Dot Product

Find the area enclosed between the curve $y = x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 19/2

b. 9

c. **9/2** ✓

d. 7/2

Area $= \int_0^3 (x)dx = [1/2x^2]_0^3 = 9/2$

43 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 5.03 Parametric Equations

State and apply Vieta's formulas to find $\sum r_i^2$ if r_1, r_2, r_3 are roots of $x^3 - 4x + 1 = 0$.

a. $\sum r_i^2 = 4$

b. $\sum r_i^2 = 16$

c. $\sum r_i^2 = 1$

d. $\sum r_i^2 = (\sum r_i)^2 - 2\sum_{i<j} r_i r_j = 0 - 2(-4) = 8$ ✓

Vieta: $\sum r_i = 0$, $\sum_{i<j} r_i r_j = -4$. $\sum r_i^2 = 0 - 2(-4) = 8$.

44 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 6.01 Polar Coordinates and Conversions

The quadratic formula gives: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. For $a = 1, b = -6, c = 8$, find x.

a. $x = -2, -4$

b. $x = 3 \pm 1$

c. $x = 1, 8$

d. $x = 2, 4$ ✓

$x = \frac{6 \pm \sqrt{36 - 32}}{2} = \frac{6 \pm 2}{2} = 4$ or 2.

45 [Challenging] • Target: Growth models • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum $1 + 4 + 7 + \cdots + (3n - 2)$ using the formula $S = n(3n - 1)/2$. Verify for $n = 4$.

a. $S_4 = 24$

b. $S_4 = 28$

c. $S_4 = 22$ ✓

d. $S_4 = 16$

$S_4 = 4(12 - 1)/2 = 4 \cdot 11/2 = 22$. Check: $1 + 4 + 7 + 10 = 22$. ✓.

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